
GreenStep Employee

User Experience and Interface Design Specification

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1. User Assumptions

GreenStep Employee serves two distinct user roles. The **Regular Employee** is the primary user who logs sustainable actions daily, tracks personal progress, joins teams, and competes on leaderboards. The **Administrator** is a designated sustainability coordinator who creates and manages challenges, reviews team requests, and exports participation data for reporting purposes.

Users are assumed to have basic digital literacy sufficient to navigate a web browser, create an online account, and interact with standard interface controls such as buttons, forms, and dropdowns. No advanced technical knowledge is required of either role. Administrators are expected to have slightly higher digital comfort, but no programming or system administration experience is necessary.

Users are expected to access the platform via a modern web browser on either a desktop computer or a mobile device. The primary usage pattern consists of brief, frequent sessions in which employees log one to three actions per day, often from a smartphone during or after their commute. Participation is voluntary or organizationally encouraged as part of a time-bounded sustainability initiative such as Earth Month. Users are motivated by a combination of personal interest in sustainability and gamification elements including points, levels, badges, and streaks. Challenge join codes are assumed to be distributed by the organization through internal channels such as email or a messaging platform.

New users are not expected to have prior knowledge of action categories or point values. A seven-step onboarding wizard guides each new user through setup and captures past sustainable habits before they begin logging new actions.

2. Environmental and Usage Constraints

GreenStep Employee is a web-only application built on Next.js 15. No native iOS or Android application is available in the current version. JavaScript must be enabled in the browser, and the application supports modern Chromium-based browsers (Chrome and Edge), Safari, and Firefox. Legacy browsers such as Internet Explorer are not supported. An active internet connection is required at all times, as the application has no offline mode or local caching strategy.

Authentication is managed entirely by Clerk. Administrator access is granted by setting an admin role flag in the Clerk dashboard and cannot be self-assigned by any user. All actions are self-reported with no automated verification, and the system assumes good-faith participation. The database is hosted on Neon, a serverless PostgreSQL provider, and availability is subject to Neon's service-level agreement. The application stores only the minimum necessary data: name, email, team membership, and logged actions.

Joining a challenge requires a six-character code distributed by an administrator. There is no public challenge discovery mechanism. Team creation requires administrator approval through a request-and-review workflow. The only available data export format is CSV. The application is deployed on Vercel and is optimized for time-bounded campaigns with small-to-medium organizations of tens to a few hundred concurrent users.

3. User Story Map

The backbone activities below represent the top-level goals users accomplish within GreenStep Employee. Each activity is paired with its key user stories.

Activity	Key User Stories
Authenticate	As an employee, I want to sign up or sign in so that I can access the platform and be redirected automatically to my dashboard.
Onboard	As a new employee, I want to complete a seven-step guided wizard so that my name, department, and past sustainable habits across five categories are captured before I begin.
Log Actions	As an employee, I want to browse actions by category, search by keyword, and sort by point value so that I can quickly find and confirm a relevant action to log.
Track Progress	As an employee, I want to view my points, level, progress bar, badges, streak, activity history, and a performance chart so that I can monitor and reflect on my sustainability engagement.
Compete	As an employee, I want to join a challenge using a six-character code, view team and individual leaderboards, and leave a challenge so that I can manage my competitive participation.
Team Collab	As an employee, I want to browse and join existing teams or request a new team so that my points contribute to a group score and I can compete collaboratively.
Administer	As an admin, I want to create challenges, approve or reject team requests, and export activity data as a CSV file so that I can manage the platform and report outcomes to stakeholders.

Table 1: User story map with backbone activities and key user stories for GreenStep Employee.

4. Journey and Interaction Model

GreenStep Employee supports four primary user journeys. In the **New Employee Journey**, the user visits the landing page, completes sign-up through Clerk, and is redirected to the onboarding wizard. The wizard walks the user through seven steps (Welcome, Transport, Energy, Waste, Food, Water, and Summary) before submitting and landing on the dashboard. In the **Returning Employee Journey**, the user signs in, navigates to the Log Action page, selects a category or searches for an action, views the action detail page, and confirms the submission to receive updated points on the dashboard. In the **Challenge and Team Journey**, the user clicks the Join Challenge button on the dashboard, enters a six-character code, and gains access to challenge-scoped data. From the Teams tab, the user either joins an existing team or submits a new team request that enters a pending state until an administrator approves or rejects it. The user then views ranked standings on the Leaderboard tab. In the **Administrator Journey**, the admin accesses the Admin page from a role-gated link in the header, reviews pending team requests, creates challenges with optional date ranges, and downloads CSV exports of all or per-challenge activity.

Figure 1 presents the complete interaction flow model for these journeys, created using Lucidchart. Green paths indicate the Regular Employee journey and blue paths indicate the Administrator journey.

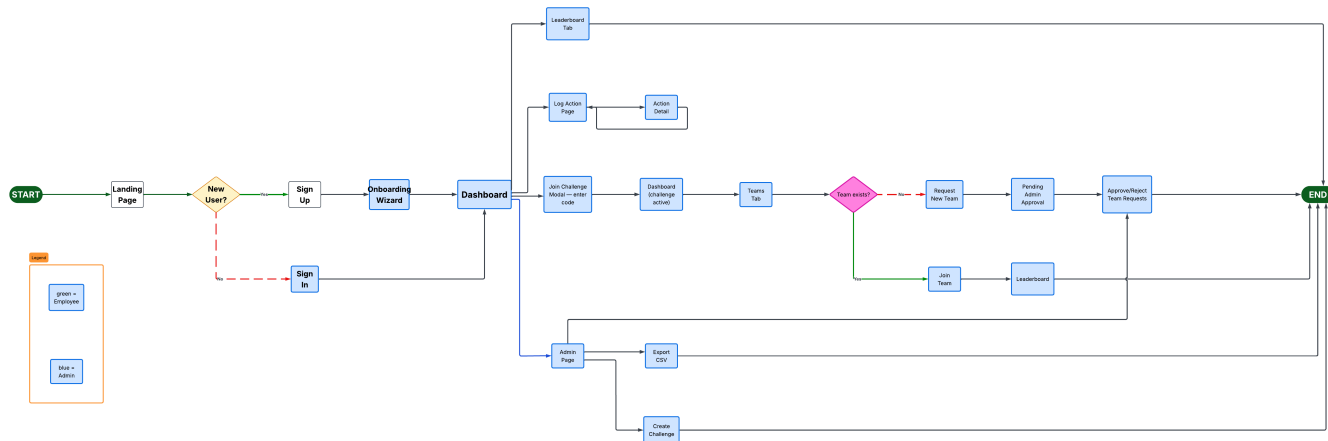


Figure 1: GreenStep Employee high-level user interaction flow model. Green paths represent the Regular Employee journey and blue paths represent the Administrator journey. Diagram created using Lucidchart.

5. Accessibility and Inclusive Design Considerations

The application is fully responsive using Tailwind CSS, with a dedicated mobile header and a fixed bottom navigation bar for small screens. All interactive controls meet the 44-pixel minimum touch target height through the `min-h-11` and `min-h-12` utility classes, ensuring usability on touchscreen devices.

The animated point counter checks the `prefers-reduced-motion` media query before applying any transition. Users who have enabled reduced motion at the operating system level receive static values instead, satisfying WCAG 2.1 Success Criterion 2.3.3 and accommodating users with vestibular disorders or motion sensitivity. All interactive elements carry `focus-visible` CSS outlines to support keyboard-only navigation. Modal dialogs include clearly labeled close buttons that are reachable via the Tab key. Icon-only controls, such as the modal close button, include `aria-label` attributes for screen reader users. Clerk’s authentication components provide accessible labeling on all sign-in and sign-up flows by default.

The primary brand color of dark forest green (`#0b5d1e`) on a white background exceeds the WCAG 2.1 AA minimum contrast ratio of 4.5 to 1 for normal text. Information is never conveyed by color alone. Badge states, challenge category pills, and status indicators always include accompanying textual labels so that users with color vision deficiencies receive the same information. All user-facing copy uses plain, jargon-free language. The onboarding wizard applies progressive disclosure by presenting one action category per step rather than displaying the full action catalog at once, reducing cognitive load for new users. Inline error messages are placed adjacent to the relevant input field for clarity.

The action catalog spans low-effort entries worth 10 points to high-effort entries worth 50 points across five categories, ensuring that employees with different lifestyles, living situations, and physical abilities can participate meaningfully. Semantic HTML5 elements including `header`, `main`, `nav`, `section`, and `button` provide structural landmarks for screen readers and other assistive technologies.

5. References

- [1] W3C, *Web Content Accessibility Guidelines (WCAG) 2.1*, Jun. 2018. [Online]. Available: <https://www.w3.org/TR/WCAG21/>
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